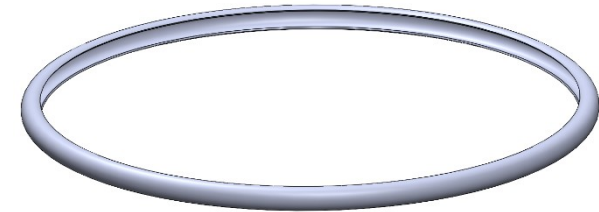


INCONEL 718 SEAL & SETUP

Present Status of the Project

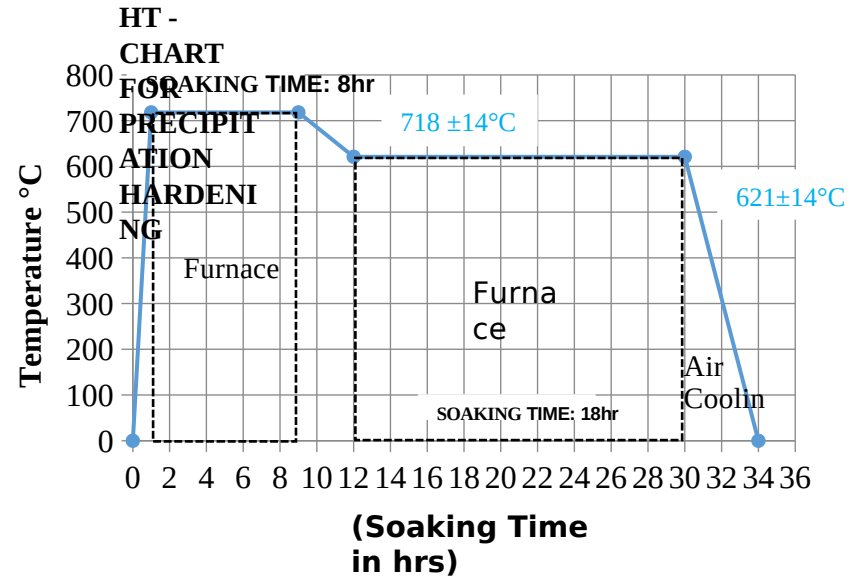
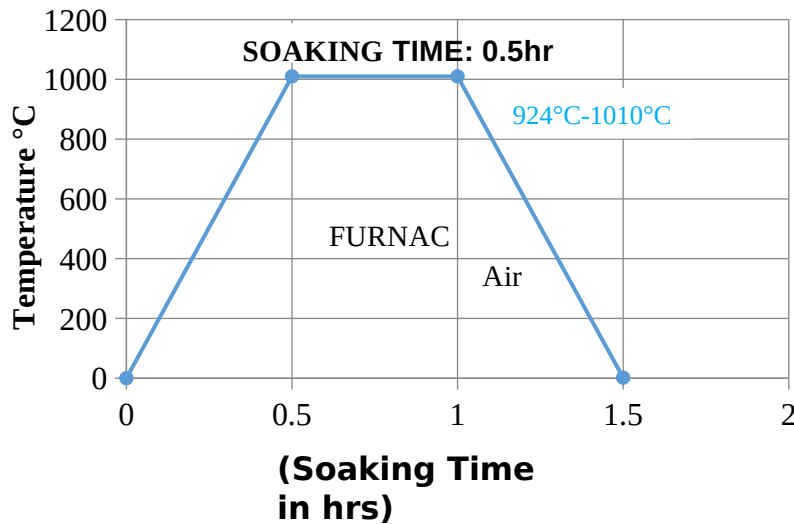
SL no	DESCRIPTION	STATUS
1	Procurement of INCONEL 718 tube and its testing.	Completed
2	3D Modelling of Small & Large-scale Setup Assembly.	Completed
3	FEM Analysis	Completed
4	Identification of vendor for custom seal development. Yet to be finalized	completed



INCONEL 718 SEAL & SETUP

The Procured Raw Material in Tube form was Heat Treated and tested :

HT - CHART FOR SOLUTION TREATMENT



Test name	Condition	Specimen Dimension in mm	Temp (°C)	Required Yield Strength (Mpa)	Resulted Yield strength (Mpa)
Tensile	Heat treated	OD (10) × T (1) × L (250)	RT	1034	556
			350	>1100	435

INCONEL 718 SEAL & SETUP

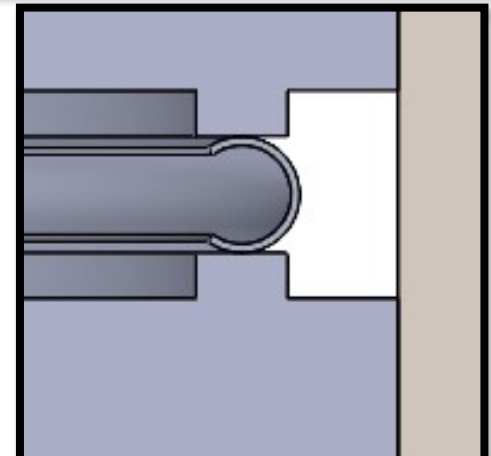
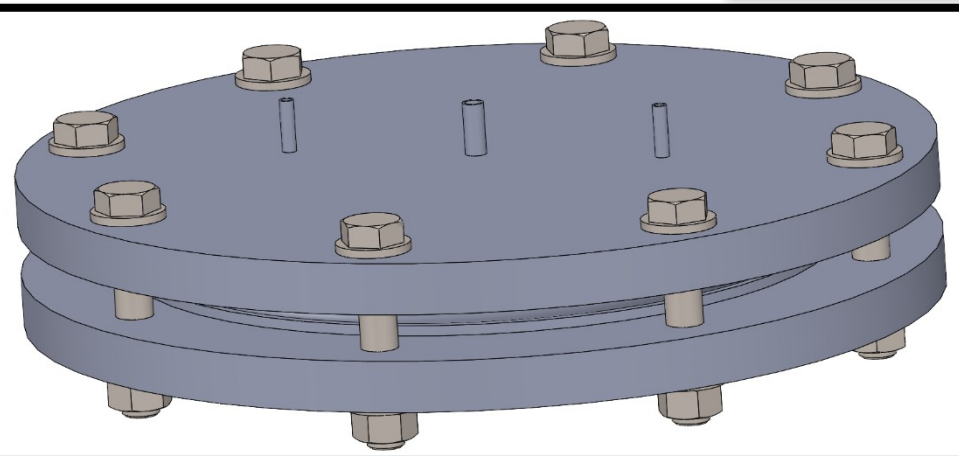
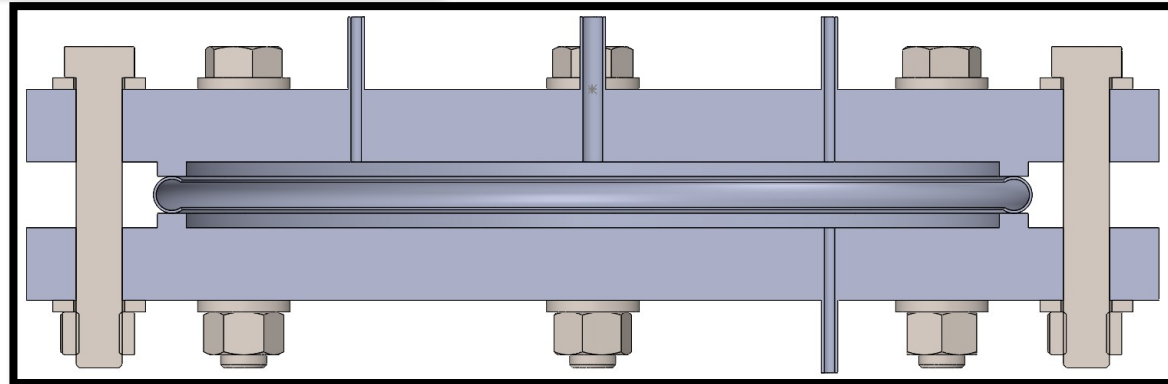
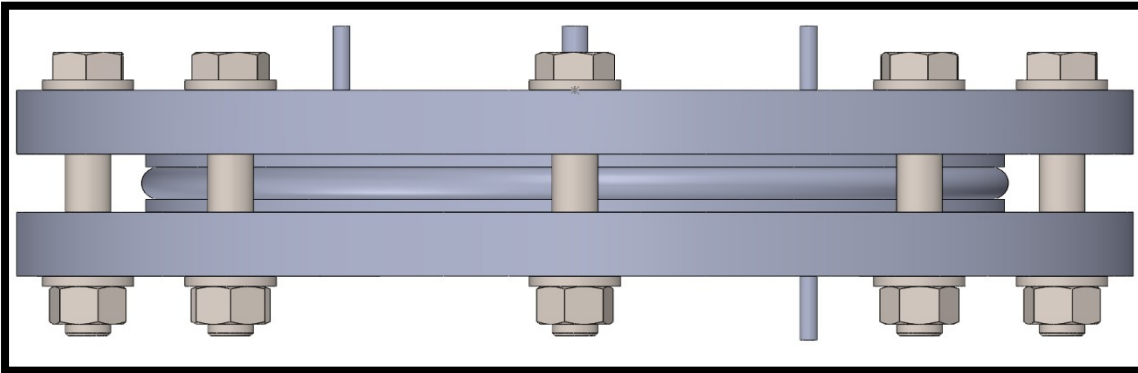
The Tube material was tested to find the Chemical Composition:

Element	SEM-EDS weight (%)	OES weight (%)	INCONEL 625 (%)	Comment	INCONEL 718 (%)	Comment
Al	0.43	0.280	0.4 max	✓	0.20-0.80	✓
Si	0.25	0.270	0.5 max	✓	0.35 max	✓
Ti	0.19	0.200	0.4 max	✓	0.65-1.15	Less
Cr	20.413333	20.200	20-23	✓	17.00-21.00	✓
Mn	0.3166667	0.140	0.5 max	✓	0.35 max	✓
Fe	4.55	5.210	5 max	✓	11-22	Less
Ni	64.89	63.500	58 min	✓	50-55	More
Nb	2.17	2.160	3.15-4.15	less	4.75-5.50	Less
Mo	6.7933333	7.590	8-10	✓	2.80-3.30	More

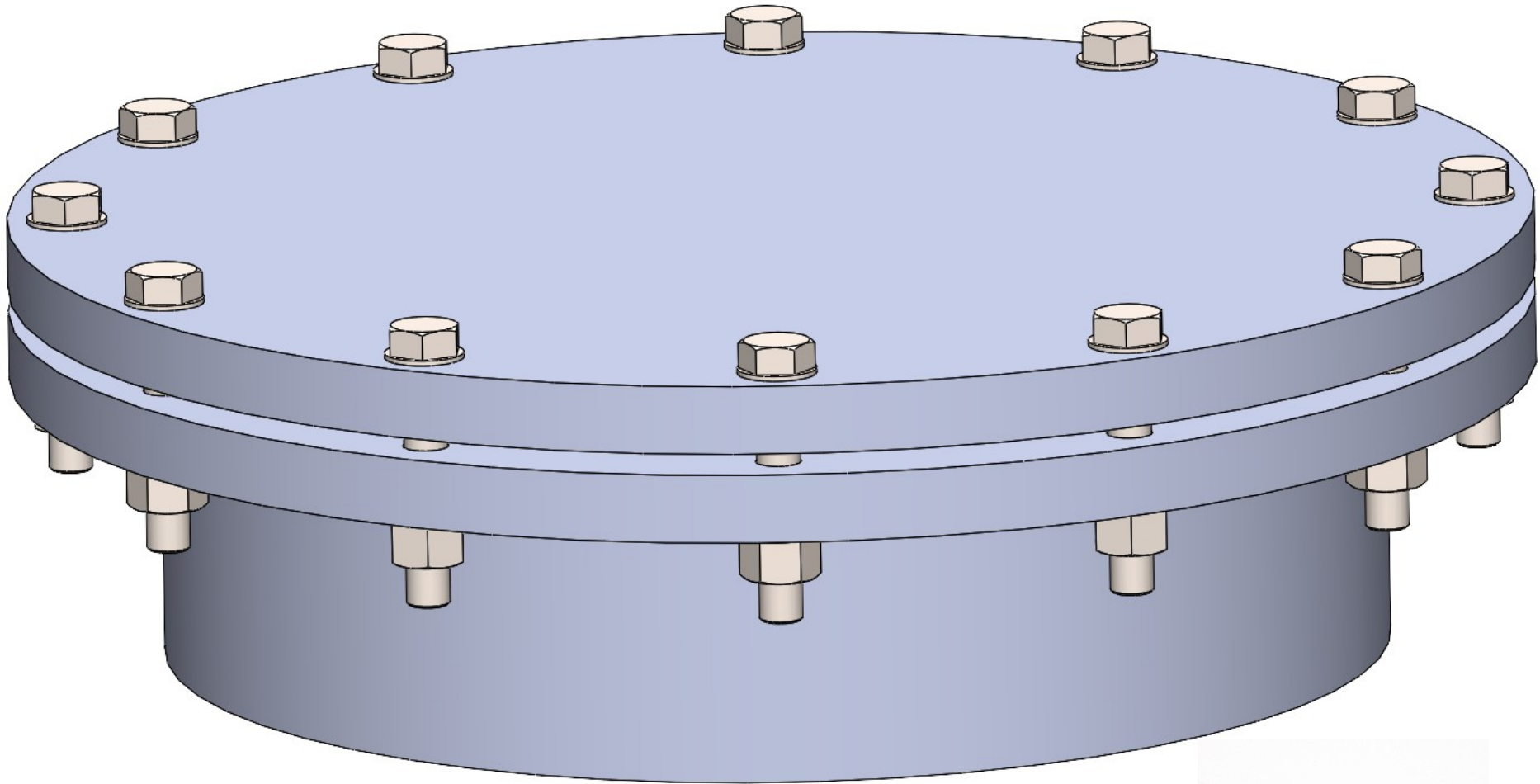


Hence, the tested sample of **INCONEL 718** favours to prove its chemical composition more towards **INCONEL 625**

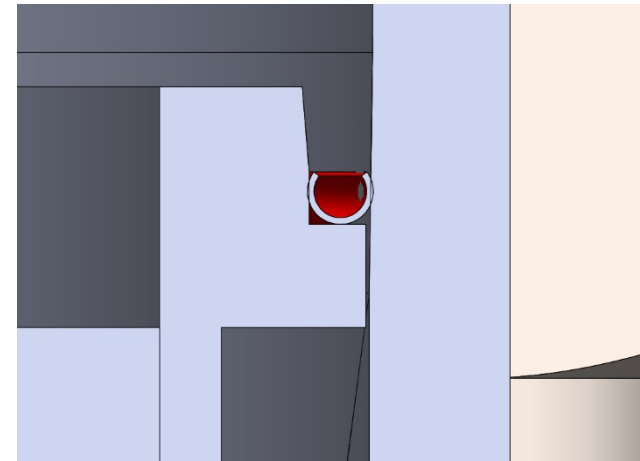
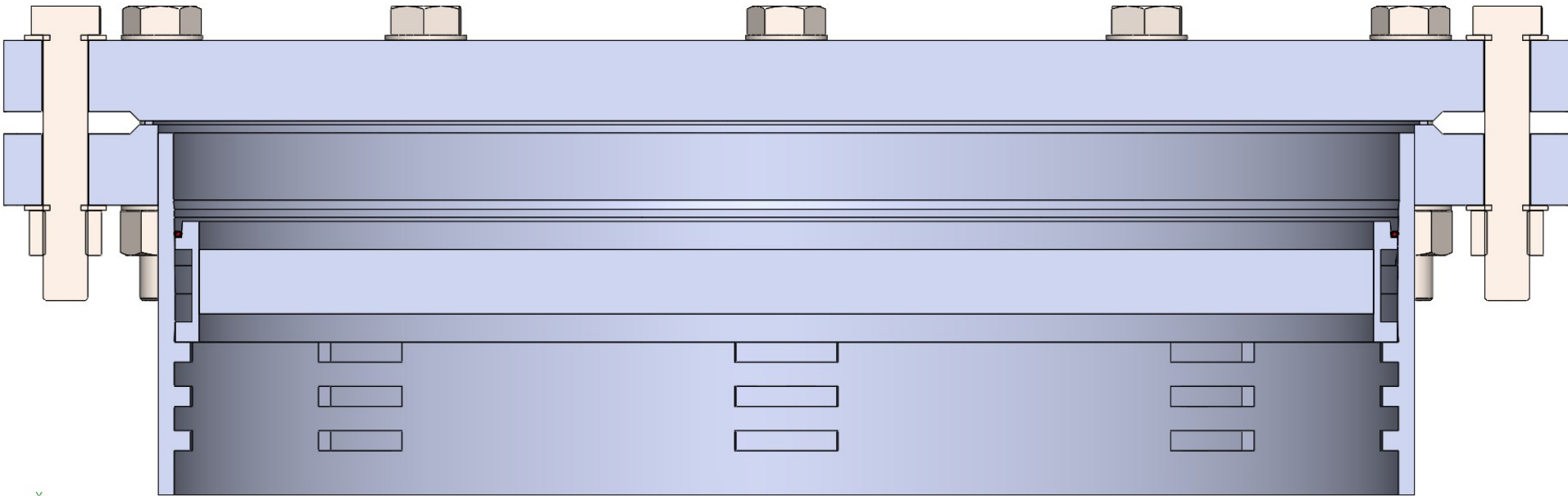
SMALL-SCALE TESTING SETUP



LARGE-SCALE TESTING SETUP

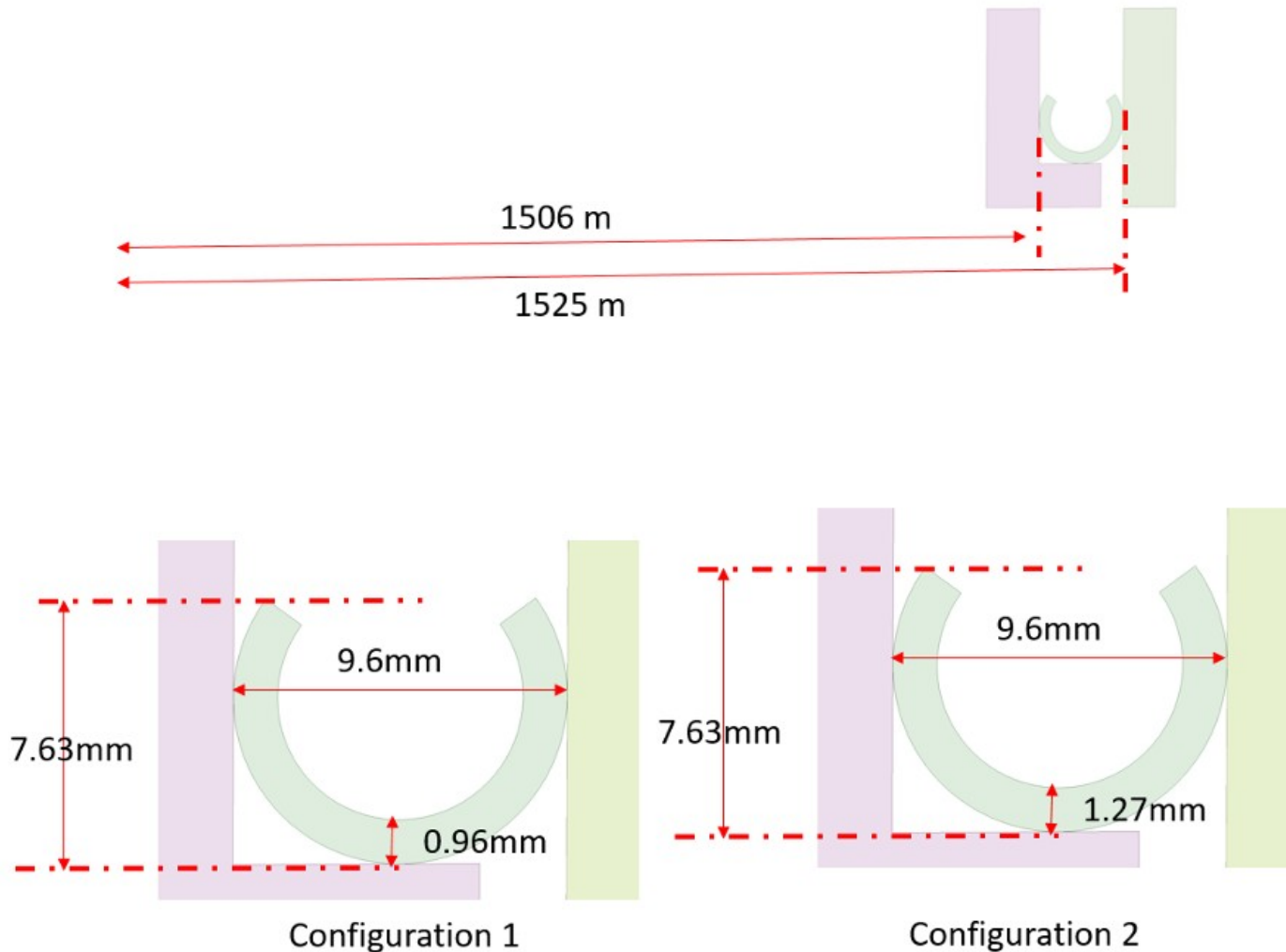


SECTION VIEW OF LARGE-SCALE SETUP



FEM FOR DESIGN & DEVELOPMENT OF INCONEL SEAL

Modelling of Seal



FEM FOR DESIGN & DEVELOPMENT OF INCONEL SEAL

Technical Details for Metal C Ring

Sl no	Seal	Parameters	Details
1		Type	Metal C ring with axial seal
2	Cavity details	Bore Diameter	1525.5(0.00/0.15) mm
3		Rod Diameter	1506(0.00/-0.15) mm
4		Nominal Cross-section	9.6 mm
5	Seal Parameters	Seal Thickness	T1=0.96 mm & T2 = 1.27 mm
6		Material	Inconel 718 AMS 5596
7		HT Condition	Cold formed+SA+PH
8		Working pressure	Max. 10 bar
9		Working	PT 0.0001-0.0002

FEM FOR DESIGN & DEVELOPMENT OF INCONEL SEAL

Summa

Sl no .	Loading condition (Thermal loads corresponding to)	Internal pressure applied (Bar)	Pre-stress applied (mm)	Thickne ss of Seal (mm)	Wall Conta ct stress (MPa)	Maxim um Stress (MPa)	Materi al Yield (MPa)	FO S
1	200°C (1524 mm)	10	0.5	0.96	661.06	909.46	1100	1.2
2	350°C (1523 mm)		1		1354	1473.3		0.74
3	200°C (1524 mm)		0.5	1.27	859.87	955.95		1.15
4	350°C (1523 mm)		1		1560.4	1523.4		0.72

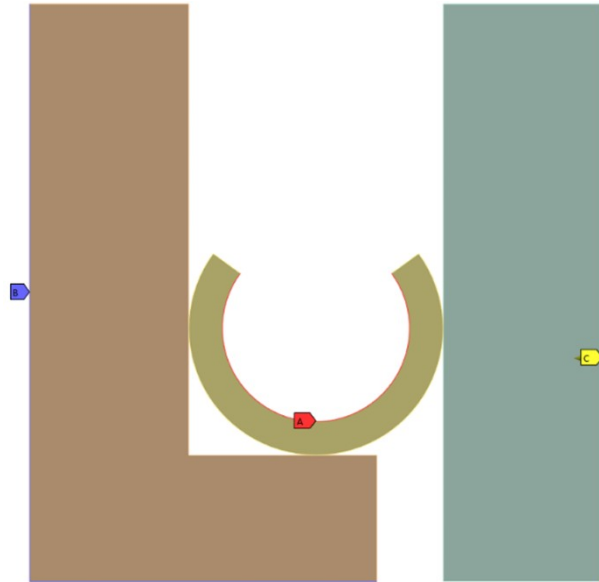
FEM FOR DESIGN & DEVELOPMENT OF INCONEL SEAL

Material Properties

Sl no.	Material used location	Material used	Density (Kg/m ³)	Yield strength (MPa)	Ultimate tensile strength (MPa)
1	Walls	Structural steel	7850	650	800
2	Seal	Inconel 718	8200	1100	1400



FEM FOR DESIGN & DEVELOPMENT OF INCONEL SEAL



A – Pressure 1 Mpa

B – Fixed Support

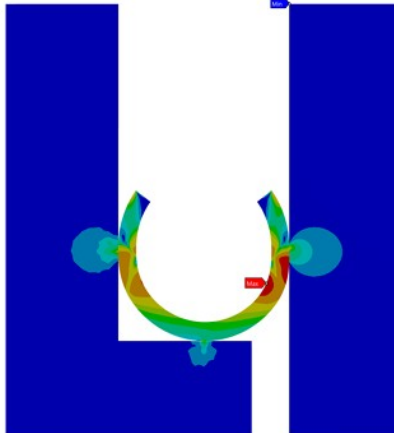
C – Displacement

Sl · n o	Parameters	Details
1	Material Assignments	- Fixed Support and Wall: Structural Steel - Seal: Inconel 718
2	Symmetry	A symmetry region setting was applied to the seal with the symmetry normal to the Y-axis.
3	Contact Regions	Friction coefficient of 0.2 between the wall-seal and fixed support-seal.
4	Boundary Conditions	Fully constrained in all directions, on two edges of the geometry.
5	Pressure	Pressure of 1 MPa was applied to the inside edge of the seal.
6	Displacement Cases	0.5 mm to the Wall. 1 mm to the Wall.
7	Thickness Cases	- seal thickness-0.96 mm. - seal thickness-1.27 mm.
8	Analysis Settings	- Analysis Type: The analysis was performed in quasi-static mode, assuming slow and steady loading conditions. - Steps: The number of steps was set to 1. - Time Settings: - Initial time step: 0.1 seconds. - Minimum time step: 0.01 seconds. - Maximum time step: 1 second.

FEM FOR DESIGN & DEVELOPMENT OF INCONEL SEAL

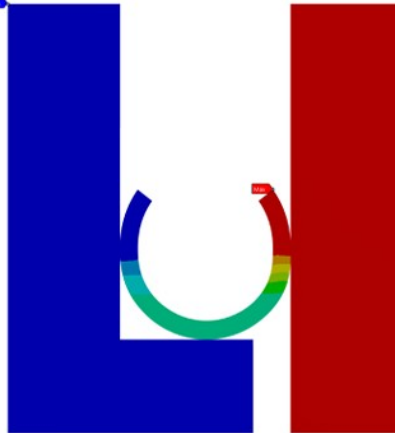
Case 1: Seal Thickness 0.96 mm and Displacement 0.5 mm

1. 0.5mm Without Spring
Equivalent Stress (1)
Type: Equivalent (von Mises) Stress
Unit: Mpa
Time: 1.1
0.00000000 1.136



909.46 Max
800.41
707.36
606.21
505.06
403.91
302.76
201.61
100.46
0.00000000 Min

1. 0.5mm Without Spring
Total Deformation
Type: Total Deformation
Unit: mm
Time: 1.1
0.00000000 1.137

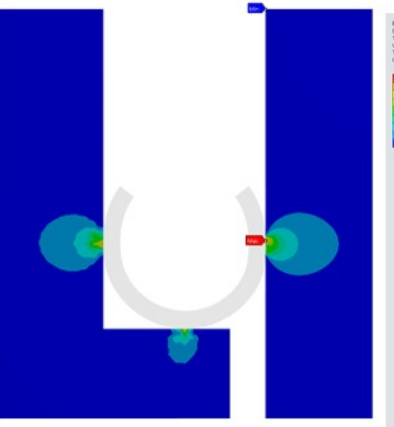


0.52121 Max
0.46122
0.40123
0.34124
0.28125
0.22126
0.16127
0.10128
0.04129
0.00000000 Min

Equivalent stress
3
Type: Equivalent
(Von-
mises)
Stress – **909.46**
Mpa

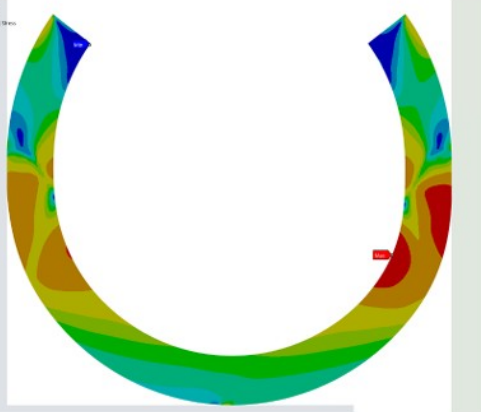
Total
Deformation –
0.52121

2. 0.5mm Without Spring
Equivalent Stress (2)
Type: Equivalent (von Mises) Stress
Unit: Mpa
Time: 1.1
0.00000000 1.137



909.46 Max
800.41
707.36
606.21
505.06
403.91
302.76
201.61
100.46
0.00000000 Min

2. 0.5mm Without Spring
Equivalent Stress
Type: Equivalent (von Mises) Stress
Unit: Mpa
Time: 1.1
0.00000000 1.137



909.46 Max
800.41
707.36
606.21
505.06
403.91
302.76
201.61
100.46
0.00000000 Min

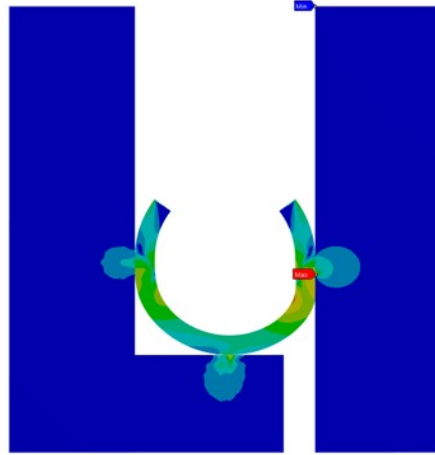
Equivalent stress
2
Type: Equivalent
(Von-
Mises)
Stress – **909.46**
Mpa

Equivalent stress
Type: Equivalent
(Von-
mises)
Stress – **909.46**
Mpa

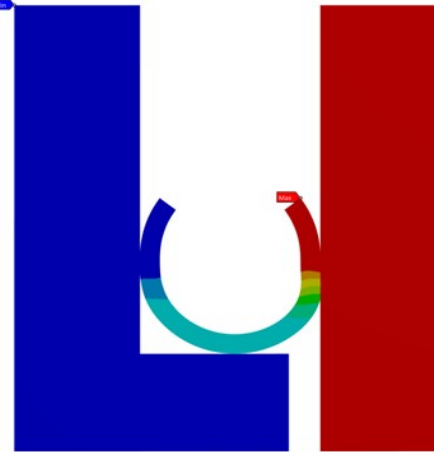
FEM FOR DESIGN & DEVELOPMENT OF INCONEL SEAL

Case 2: Seal Thickness 0.96 mm and Displace

NI: Tens Without Spring
Equivalent Stress 3
Type: Equivalent (von Mises) Stress
Unit: MPa
Time: 1 s
07-05-2023 12:13
1473.3 Max
1208.6
1145.9
982.19
818.48
655.3
491.7
328.11
164.31
0.0000029 Min

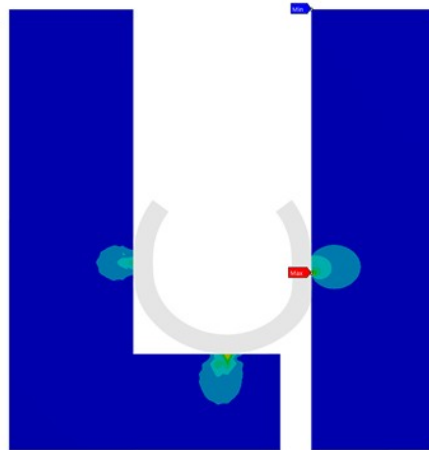


NI: Tens Without Spring
Total Deformation
Type: Total Deformation
Unit: mm
Time: 1 s
07-05-2023 12:13
1.0274 Max
0.97123
0.79908
0.64862
0.51977
0.43662
0.38246
0.32811
0.17475
0 Min

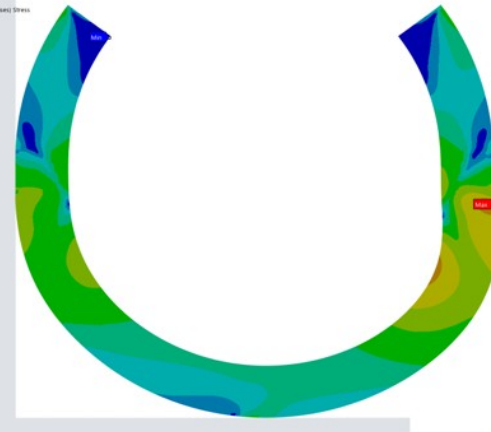


Total Deformation – **1.0274mm**

NI: Tens Without Spring
Equivalent Stress 2
Type: Equivalent (von Mises) Stress
Unit: MPa
Time: 1 s
07-05-2023 12:16
1354 Max
1208.6
1053.1
982.67
792.28
657.78
491.34
328.09
164.45
0.0000029 Min



NI: Tens Without Spring
Equivalent Stress
Type: Equivalent (von Mises) Stress
Unit: MPa
Time: 1 s
07-05-2023 12:17
1473.3 Max
1208.7
1145.1
982.1
818.9
655.3
491.7
328.11
164.31
0.01203 Min



Equivalent stress
Type: Equivalent (Von-mises)
Stress – **1473.3 Mpa**

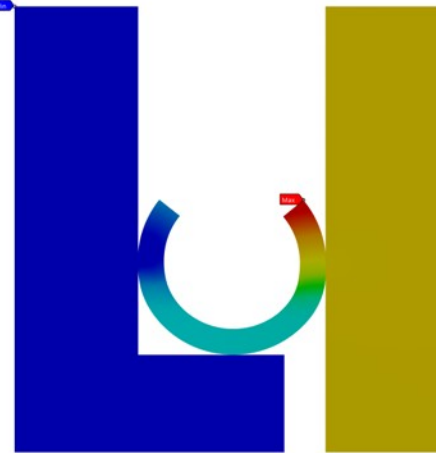
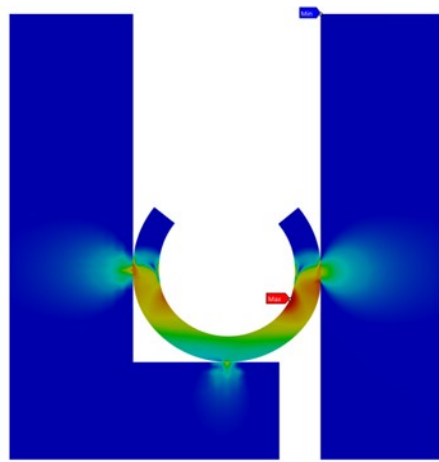
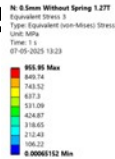
Equivalent stress
3
Type: Equivalent (Von-mises)
Stress – **1473.3 Mpa**

Equivalent stress
2
Type: Equivalent (Von-Mises)
Stress – **1354 Mpa**

FEM FOR DESIGN & DEVELOPMENT OF INCONEL SEAL

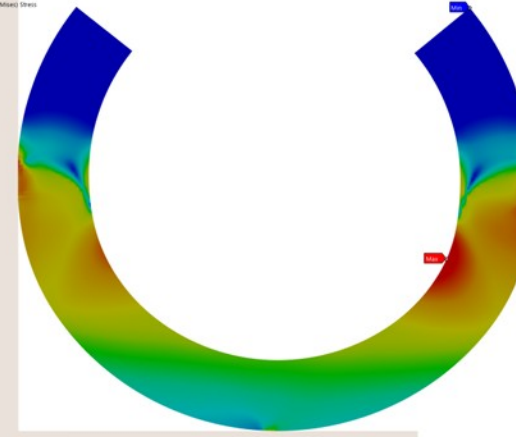
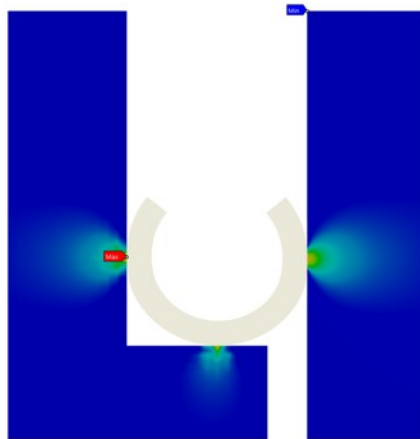
Case 3: Seal Thickness 1.27 mm and Displace

Equivalent stress
3
Type: Equivalent
(Von-mises)
Stress - **955.95 Mpa**



Total
Deformation -
0.6567 mm

Equivalent stress
2
Type: Equivalent
(Von-Mises)
Stress - **859.87 Mpa**

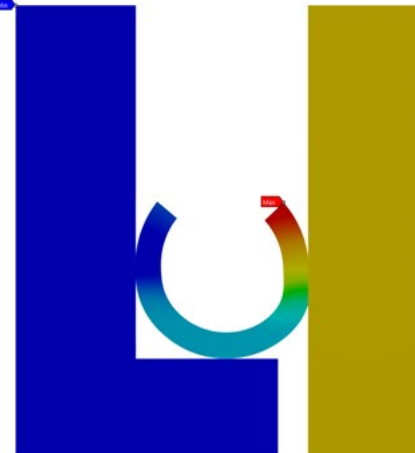
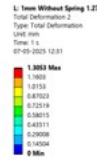
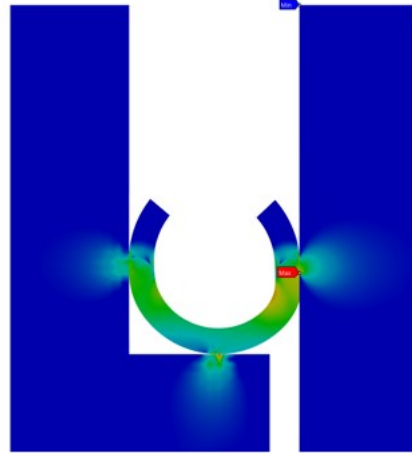
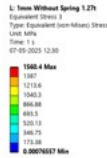


Equivalent stress
Type: Equivalent
(Von-mises)
Stress - **955.95 Mpa**

FEM FOR DESIGN & DEVELOPMENT OF INCONEL SEAL

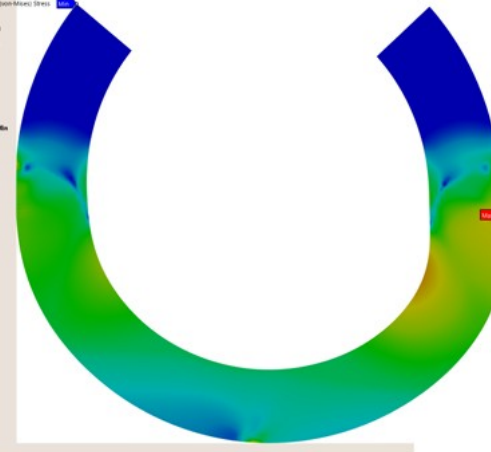
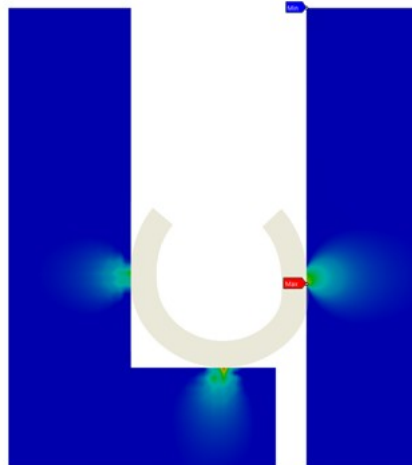
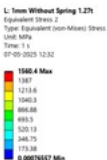
Case 3: Seal Thickness 1.27 mm and Displacement 0.5 mm

Equivalent stress
3
Type: Equivalent
(Von-mises)
Stress – **1560.4
Mpa**



Total
Deformation –
1.3053 mm

Equivalent stress
2
Type: Equivalent
(Von-Mises)
Stress – **1560.4
Mpa**



Equivalent stress
Type: Equivalent
(Von-mises)
Stress – **1523.4
Mpa**

**THANK
YOU**

INCONEL 718 SEAL & SETUP

Coordinating with INCONEL SEAL Manufacturers



Company Name - **FLEXO-RESIST**

Location - **Hadapsar, Pune,
Maharashtra**

Description - Flexo-resist is already
Manufacturer,

Supplier, Exporter of Metal Seals, Metal O
Ring
Seals.

